

Inclusive and Participatory Community Information Systems

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Project Deployed URL: <https://swe599-project-frontend.onrender.com/>

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1. Project Overview and Objectives

The main idea behind this project is to create a platform that helps communities collect, organize, and share information in a structured way. Instead of relying on chats, emails, or scattered messages where details easily get lost, this system brings everyone's contributions together in one place. When people unite for a shared goal-like rebuilding after a disaster, planning local activities, or coordinating support—they need more than just conversation. They need a clear and organized way to express what they can offer, what they need, and how they can contribute. This platform aims to make that process simpler, more efficient, and more meaningful for everyone involved.

1.1 Why Information Management Important?

In today's connected world, communities share information through many different channels—group chats, social media, emails, and even phone calls. While these tools are great for fast communication, they are not designed to keep information structured and easy to track. As a result, important details often end up scattered across different places, making it difficult to see the full picture of what a community has and what it still needs.

This problem becomes even more challenging as the number of participants grows. A small group may be able to manage updates informally, but once dozens or hundreds of people start contributing, messages quickly pile up. Valuable offers—like someone donating tools or volunteering their skills—get buried under a flood of new messages. Coordinators then have to waste time scrolling through conversations just to figure out who offered what, instead of focusing on real work.

In short, traditional communication platforms are built for talking/texting, not organizing. They cannot scale effectively when large groups need to collect structured information, and they leave organizers with the exhausting job of manually sorting through hundreds of messages/posts. This is why communities need a system that keeps information organized, searchable, and easy to understand, no matter how large the group becomes.

1.2 The Contribution of the Project on Information Management

This platform solves these problems by offering a structured way to collect information. Instead of open conversations where important details easily get lost, the system allows community organizers decide exactly what type of information they want to collect. This might include specific items (like tools or building materials), skills (such as construction experience), or general support (like volunteers or transportation).

Once the structure is defined, community members can submit their information in a clear and consistent format. The platform then organizes everything, making it easy for everyone to see what resources are available, what is still missing, and who is offering what.

2. The Important Aspects of The Proposed System

2.1 Structured Data Collection

The main advantage of our system is that it helps collect information in a structured and consistent way. Instead of letting everyone share details in whatever format they choose, community organizers can set up exactly what kind of information they want to gather. This ensures that all contributions follow the same format, which makes the data much easier to organize, search, and understand.

For example, if a community needs construction tools, the organizer can create a dropdown menu with different tool categories. When members contribute, they simply choose from these predefined options instead of typing out their own descriptions. This way, the system can automatically group and display all too related offers together, no matter when they were submitted. This saves time, reduces confusion, and makes it much easier to get a clear view of the resources available.

2.2 Scalability

The system is built to grow with the community. Even as more people join and the amount of information increases, the structured format keeps everything organized and easy to use. With features like filtering, sorting, and searching, users can quickly find what they need even in large datasets. This is something traditional, unstructured communication tools simply cannot handle once the group gets big.

3. Software Requirements Specifications

3.1 User Requirements

3.1.1 Regular User Requirements

- 3.1.1.1 The system shall allow a new user to register by providing required information (email, name, surname, username, password, profession, date of birth, and consent)
- 3.1.1.2 The system shall validate user input during registration and display inline error messages for invalid fields.
- 3.1.1.3 The system shall log in a user automatically after successful registration.
- 3.1.1.4 The system shall allow registered users to log in using a valid username/email and password.
- 3.1.1.5 The system shall deny access if credentials are invalid and display an appropriate error message.
- 3.1.1.6 The system shall allow authenticated users to log out securely.
- 3.1.1.7 The system shall allow users to view their profile information.
- 3.1.1.8 The system shall allow users to edit their email, username, profession.
- 3.1.1.9 The system shall validate edited fields.
- 3.1.1.10 The system shall allow users to change their password after verifying the current one.
- 3.1.1.11 The system shall allow users to permanently delete their account upon confirmation.

- 3.1.1.12 The system shall allow users to view other users' profiles by username.
- 3.1.1.13 The system shall allow users to search for other users by name.

3.1.2 Admin User Requirements

- 3.1.2.1 The system shall provide an admin dashboard accessible only to users with admin privileges.
- 3.1.2.2 The system shall display system statistics including total users, communities, and tabs in the admin dashboard.
- 3.1.2.3 The system shall allow admin to view all users in the system.
- 3.1.2.4 The system shall allow admin to search users by username, name, surname, or date of birth.
- 3.1.2.5 The system shall allow admin to sort user lists by username, name, surname, or date of birth.
- 3.1.2.6 The system shall allow admin to delete user accounts.
- 3.1.2.7 The system shall allow admin to view all communities in the system.
- 3.1.2.8 The system shall allow admin to search communities by title, description, or creator.
- 3.1.2.9 The system shall allow admin to sort community lists by title, creation date, or number of tabs.
- 3.1.2.10 The system shall allow admin to view all tabs in the system.
- 3.1.2.11 The system shall allow admin to search tabs by name or community ID.
- 3.1.2.12 The system shall allow admin to sort tab lists by community ID or number of inputs.
- 3.1.2.13 The system shall allow admin to perform all community management operations such as create, update, and delete.

3.2 Community Related Requirements

3.2.1 Community Requirements

- 3.2.1.1 The system shall display all available communities with title and description.
- 3.2.1.2 The system shall allow users to search communities by title or description.
- 3.2.1.3 The system shall allow users to filter search results dynamically as they type.
- 3.2.1.4 The system shall display communities created and contributed to by the user under separate tabs.
- 3.2.1.5 The system shall allow authenticated users to view community details, including contributions and discussions.
- 3.2.1.6 The system shall allow authenticated users to create new communities by providing a title, description, and custom data schema.
- 3.2.1.7 The system shall assign the community creator as the space creator.
- 3.2.1.8 The system shall allow community creators to define and modify the structure, including tabs and input types.
- 3.2.1.9 The system shall allow community creators to update the community title and description.
- 3.2.1.10 The system shall allow community creators or administrators to delete communities.
- 3.2.1.11 The system shall enforce a maximum limit of 10 tabs per community.

3.2.1.12 The system shall allow creators to organize community data into multiple tabs with custom names, colors, and descriptions.

3.2.2 Tab Requirements

3.2.2.1 The system shall allow community creators to add tabs to their communities.

3.2.2.2 The system shall allow community creators to modify tab properties (name, color, description).

3.2.2.3 The system shall allow community creators to delete tabs from their communities.

3.2.2.4 The system shall allow community creators to configure input types for each tab.

3.2.2.5 The system shall display tabs with their assigned colors for visual organization.

3.2.2.6 The system shall maintain tab display order within each community.

3.2.3 Input Type Requirements

3.2.3.1 The system shall support the following input types: free text, dropdown list, multiple select, date, URL, and location.

3.2.3.2 The system shall require dropdown and multiple select input types to have at least one option item defined.

3.2.3.3 The system shall prevent free text, date, URL, and location input types from having option items.

3.2.3.4 The system shall allow community creators to add input types to tabs during community creation or update.

3.2.3.5 The system shall allow community creators to modify input type properties (name, type, items).

3.2.3.6 The system shall allow community creators to delete input types from tabs.

3.2.3.7 The system shall validate input type configurations according to their type requirements.

3.2.4 Input Requirements

3.2.4.1 The system shall provide a date picker interface for date input types.

3.2.4.2 The system shall prevent manual text entry in date input fields.

3.2.4.3 The system shall require users to select dates using the date picker.

3.2.4.4 The system shall store date values as strings in YYYY-MM-DD format.

3.2.4.5 The system shall display selected dates when viewing or updating date inputs.

3.2.4.6 The system shall allow users to enter URLs in URL input fields.

3.2.4.7 The system shall store URL values as strings.

3.2.4.8 The system shall display URLs as clickable links in input details.

3.2.4.9 The system shall open URLs in a new browser tab when clicked.

3.2.4.10 The system shall require users to select a country first before selecting a city for location inputs.

3.2.4.11 The system shall provide a dropdown list of all countries for country selection.

3.2.4.12 The system shall fetch cities for a selected country from an external API.

3.2.4.13 The system shall display a loading indicator while fetching cities.

- 3.2.4.14 The system shall disable country selection while cities are being fetched.
- 3.2.4.15 The system shall allow users to proceed with an empty city if no cities are retrieved.
- 3.2.4.16 The system shall provide an optional detailed address field with a maximum of 500 characters.
- 3.2.4.17 The system shall store location data as a JSON string containing country, city, and address.
- 3.2.4.18 The system shall parse and display location data showing country, city, and detailed address in input details.
- 3.2.4.19 The system shall allow users to update location inputs with pre-filled country, city, and address values.

3.2.5 Contribution Requirements

- 3.2.5.1 The system shall allow members to submit structured contributions based on a community's schema.
- 3.2.5.2 The system shall allow authenticated users to submit inputs (contributions) to community tabs based on the tab's input structure.
- 3.2.5.3 The system shall validate inputs against the tab's input structure and field formats.
- 3.2.5.4 The system shall require all mandatory fields to be filled before submission.
- 3.2.5.5 The system shall allow users to view all inputs within a community tab.
- 3.2.5.6 The system shall display input details, including creator name, contact information, creation date, and all input field values.
- 3.2.5.7 The system shall allow users to update their own inputs.
- 3.2.5.8 The system shall allow users to delete their own inputs.
- 3.2.5.9 The system shall prevent users from updating or deleting inputs created by other users.
- 3.2.5.10 The system shall allow users to search inputs by creator name or any input field value.
- 3.2.5.11 The system shall allow users to filter inputs using advanced filter options.
- 3.2.5.12 The system shall allow users to sort inputs by creation date in ascending or descending order.

3.2.6 Search and Filter Requirements

- 3.2.6.1 The system shall provide a search box for filtering inputs within a community tab.
- 3.2.6.2 The system shall search across creator names and all input field values, including nested items and parsed location data.
- 3.2.6.3 The system shall provide an advanced filter dialog for filtering inputs by specific criteria.
- 3.2.6.4 The system shall allow filtering dropdown and multiselect inputs by selecting desired values from available options.
- 3.2.6.5 The system shall allow filtering free-text inputs using text-based search.
- 3.2.6.6 The system shall allow filtering date inputs by selecting a date to show inputs with dates earlier than the selected date.

- 3.2.6.7 The system shall allow filtering location inputs by selecting a country or a country and city from unique existing entries.
- 3.2.6.8 The system shall display active filters as removable tags in the filter dialog.
- 3.2.6.9 The system shall apply all selected filters simultaneously using AND logic.
- 3.2.6.10 The system shall allow users to remove individual filters or clear all filters.
- 3.2.6.11 The system shall update the filter button text to "Clear Filter" when filters are active.
- 3.2.6.12 The system shall exclude URL input types from filter options.

3.3 System Requirements

3.3.1 AI-Powered Feature Requirements

- 3.3.1.1 The system shall allow users to generate an AI-based summary of discussions and contributions in a community.
- 3.3.1.2 The system shall send community content to an external AI API and display the returned summary text.

3.3.2 Navigation Requirements

- 3.3.2.1 The system shall provide a consistent navigation menu accessible throughout the application.
- 3.3.2.2 The system shall include links to "Profile," "My Communities," "Create Community," "All Users" (for authenticated users), and "Logout."
- 3.3.2.3 The system shall include an "Admin Dashboard" link for users with admin privileges.
- 3.3.2.4 The system shall prevent navigation to authenticated pages after a user logs out.

3.3.3 Security Requirements

- 3.3.3.1 The system shall hash all passwords using bcrypt or an equivalent secure algorithm.
- 3.3.3.2 The system shall never store passwords in plain text.
- 3.3.3.3 The system shall generate secure JWT tokens for each login session.
- 3.3.3.4 The system shall require authentication tokens for all protected API endpoints.
- 3.3.3.5 The system shall verify user permissions before allowing community updates or deletions.
- 3.3.3.6 The system shall verify user permissions before allowing input updates or deletions.
- 3.3.3.7 The system shall allow password reset only through secure, token-based verification.
- 3.3.3.8 The system shall prevent unauthorized access to admin functions.

4. Status of the Project

In the second week of November, after a meeting with my instructor, Suzan Üsküdarlı, I received the first information about the project's goals, objectives, and structure. Based on what I learned, I started to confirm whether my understanding was correct and tried to make the abstract ideas more

concrete. First, I began forming the software requirements in my mind. Then I started designing general page mock-ups. For this purpose, I created a project file in Figma and visualized all the pages there.

Next, I started working on the coding and infrastructure of the main flows, such as authentication and authorization. I coded the basic pages like Login, Register, Profile Page, Update Profile, and Delete Profile using the design mock-ups as guidance. Later, I shared my designs, the pages I coded, and my software requirements specification analysis with Suzan Üsküdarlı in another meeting. She confirmed that I was on the right track, but also gave feedback on areas that could be improved. I focused on these points, updated the existing flows and logic, and created the designs and SRS documents for the missing parts. Since one of the main goals of the project was structured information gathering, I started designing different possible models and relations. In the later stages, I integrated the data-handling mechanism I designed into the project in a dynamic and flexible way. I created the general structure of the “Community” feature, added the related database models and API endpoints, and made the CRUD operations work. I also explored different ideas about how to store input-gathering data in the database and decided to use a JSON structure as the most suitable option. Towards the final stages, based on the feedback from our last meeting, I integrated a more advanced search and filtering feature. I also expanded the data-management part by adding different input types to the forms inside communities. After this, I added unit tests for all existing endpoints and created a script to run them together, testing the entire system. Once I fixed the issues, I moved to deployment. I deployed the database, frontend, and backend separately on Render and confirmed that the final version was working.

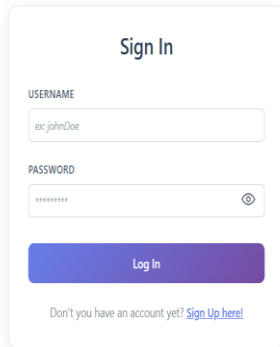
To summarize: I built a system that collects user inputs in a structured way, displays them in a structured way in the UI, and stores them in a structured way in the database. I kept the project dynamic and resistant to change, made it accessible through deployment, and added unit tests to make it easier to manage. On GitHub, I created tasks with detailed explanations and linked them to milestones so the project history can be easily followed. With the documentation I kept in parallel, I also explained the meaning and purpose of each event in the project.

5. Status of the Deployment

I deployed the project as three separate parts: database, frontend, and backend. I used the Render platform to deploy each part publicly. First, I deployed the database and obtained the connection string. Then, using the GitHub repository URL, I deployed the backend as a web service and set the database connection string as an environment variable. Finally, during the frontend deployment, I added the backend URL as an environment variable and connected all parts together. I accessed the project using the frontend URL and ran the related tests. The project is currently live and running on a public URL.

6. Design Documents

6.1 Application Screens



Sign In

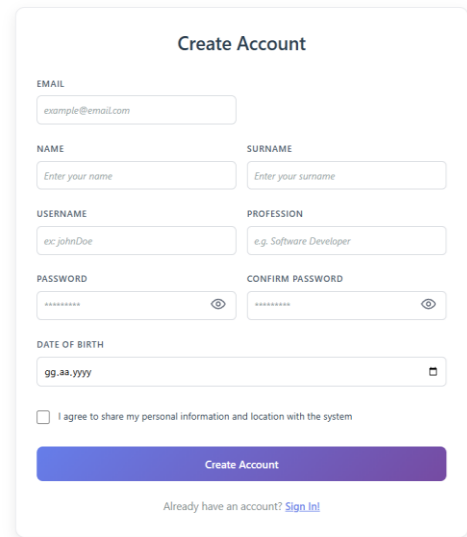
USERNAME
ex: johnDoe

PASSWORD
[masked]

Log In

Don't you have an account yet? [Sign Up here!](#)

Login



Create Account

EMAIL
example@email.com

NAME
Enter your name

SURNAME
Enter your surname

USERNAME
ex: johnDoe

PROFESSION
e.g. Software Developer

PASSWORD
[masked]

CONFIRM PASSWORD
[masked]

DATE OF BIRTH
gg.aa.yyyy

☐ I agree to share my personal information and location with the system

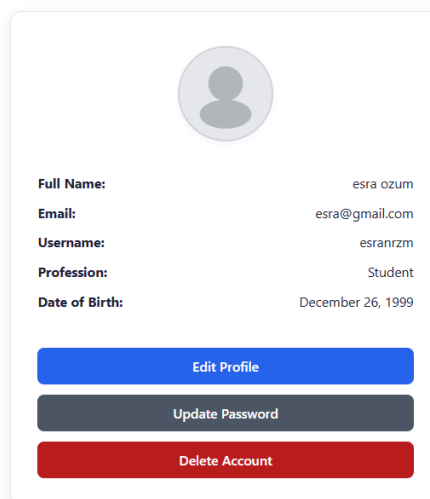
Create Account

Already have an account? [Sign In!](#)

Register



My App [Home](#) [ComHub Users](#) [Create Community](#) [User Profile](#)



[Profile Picture]

Full Name: esra ozum

Email: esra@gmail.com

Username: esranrzm

Profession: Student

Date of Birth: December 26, 1999

Edit Profile

Update Password

Delete Account

Profile Screen

Welcome to My App

You have successfully logged in! This is your main dashboard.

Search communities...

Kadın Kooperatifleri Dayanışma ve Gelişim Ağı

Kadın kooperatiflerinin üretimden pazarlamaya, finansmandan eğitim ihtiyaçlarına kadar tüm süreçlerini daha görünür ve sürdürülebilir hale getirmek için buradayız. Bu topluluk; deneyim paylaşımı, ihti...

Opportunity Classroom: Linking Graduates with Students in Need

Join a community where your skills can change a life. If you're a new teacher struggling to find opportunities, here you can gain real experience while helping a student who can't afford private lessos...

sample

sample

Children in Van need a new school building

Our school in Van has been badly damaged by recent winter storms. The old building is no longer safe or healthy for students, with a weak structure and poor insulation. We urgently need the community ...

new title2

new description

Home Screen

My App

Home

ComHub UsersCreate Community

Opportunity Classroom: Linking Graduates with Students in Need

Join a community where your skills can change a life. If you're a new teacher struggling to find opportunities, here you can gain real experience while helping a student who can't afford private lessons. Create your profile, share what you can teach, and make a difference. One lesson from you could be the support a child has been waiting for. Let's build brighter futures together. let's open doors, lift each other up, and change futures one lesson at a time.

Community Details

COMMUNITY OWNER
[esra ozum](#)
CREATED AT
11-12-2025
NUMBER OF INPUTS
3

Update Community

Delete Community

Mathematics (for teachers)

Physics (for teachers)

Chemistry (for teachers)

Music (for teachers)

"I Need Private Lessons" (for students)

Any comments about

A dedicated space for math teachers to connect with students who need support. Share your skills, build experience, and help make math clearer and more accessible for every learner.

Community Inputs

Filter

Search inputs...

Add Input to tab Mathematics (for teachers)

CREATED BY Sort	CREATED AT Sort ↓	DETAILS	ACTIONS
emre ozum	11-12-2025	See details	-

Community Details Screen

My App [Home](#) [ComHub Users](#) [Create Community](#) [User Profile](#)

Opportunity Classroom

Join a community where your skills can change the world. Gain real experience while helping a student who needs it. One lesson from you could be the difference. Open doors, lift each other up, and change the world.

Community Details

COMMUNITY OWNER: [A. Ozum](#)

CREATED AT: 12-2025

NUMBER OF INPUTS: 0

[Update Community](#) [Delete Community](#)

Community Inputs

Mathematics (for teachers) **Physics (for teachers)**

A dedicated space for physics teachers to connect and share their knowledge.

Community Inputs

CREATED BY: [Sort](#)

Fill form to add the input

What was your GPA? (0 / 200)

Enter details...

Do you have any experience on teaching?

Select option

How old are you?

Select option

Gender

Select option

Where would you prefer to do the lessons?

[Cancel](#) [Add](#)

Add Community Input Dialog

My App [Home](#) [ComHub Users](#) [Create Community](#) [User Profile](#)

Create Community

Community Title

Ex: Children in Van need a new school building

0 / 200

Community Description

Ex: Children who lives Van Agartı need a new school building with new equipments...

0 / 500

Create Community Screen 1

My App

Home

ComHub Users
Create Community

464 / 500

Community Input Tabs

Update tabs to organize different types of community inputs. Each tab can have its own input types.

Mathematics (for teachers) 9 inputs

Tab Name

Mathematics (for teachers)

Tab Color

Tab Description (Optional)

A dedicated space for math teachers to connect with students who need support. Share your skills, build experience, and help make math clearer and more accessible for every learner.

A dedicated space for math teachers to connect with students who need support. Share your skills, build experience, and help make math clearer and more accessible for every learner.

When did you graduate? (year) (Free Text)

Edit

Where did you study? (university name) (Free Text)

Edit

What was your cGPA? (Free Text)

Edit

Do you have any experience on teaching? (Dropdown)

Edit

Ex: 0-1 year

Edit

Ex: 2-3 year

Edit

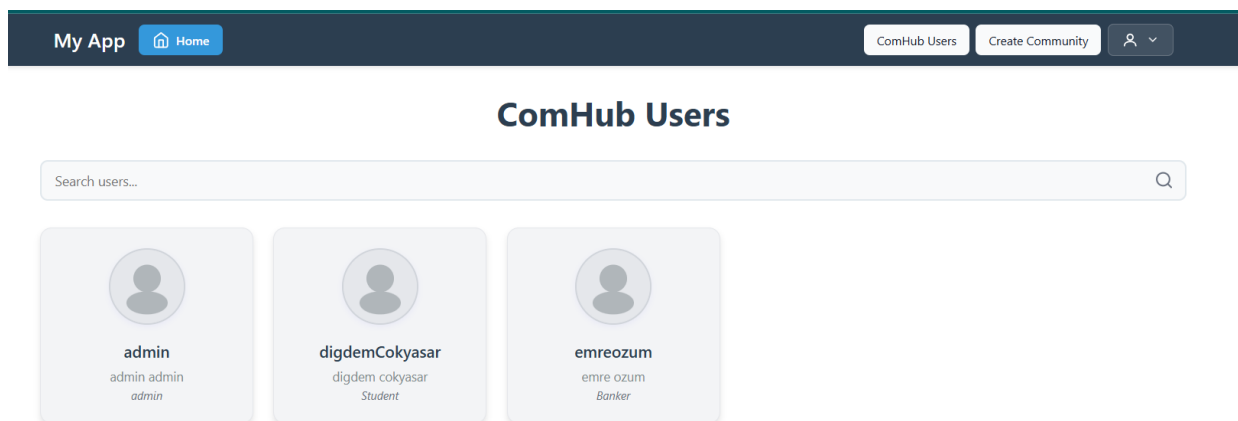
Ex: 4-5 year

Edit

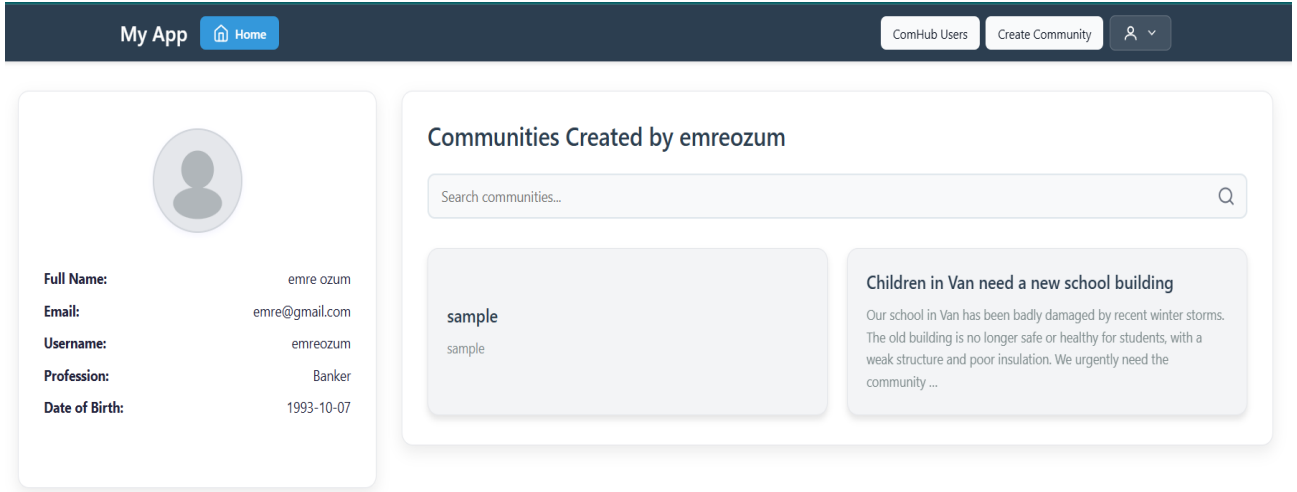
Ex: 5+

Edit

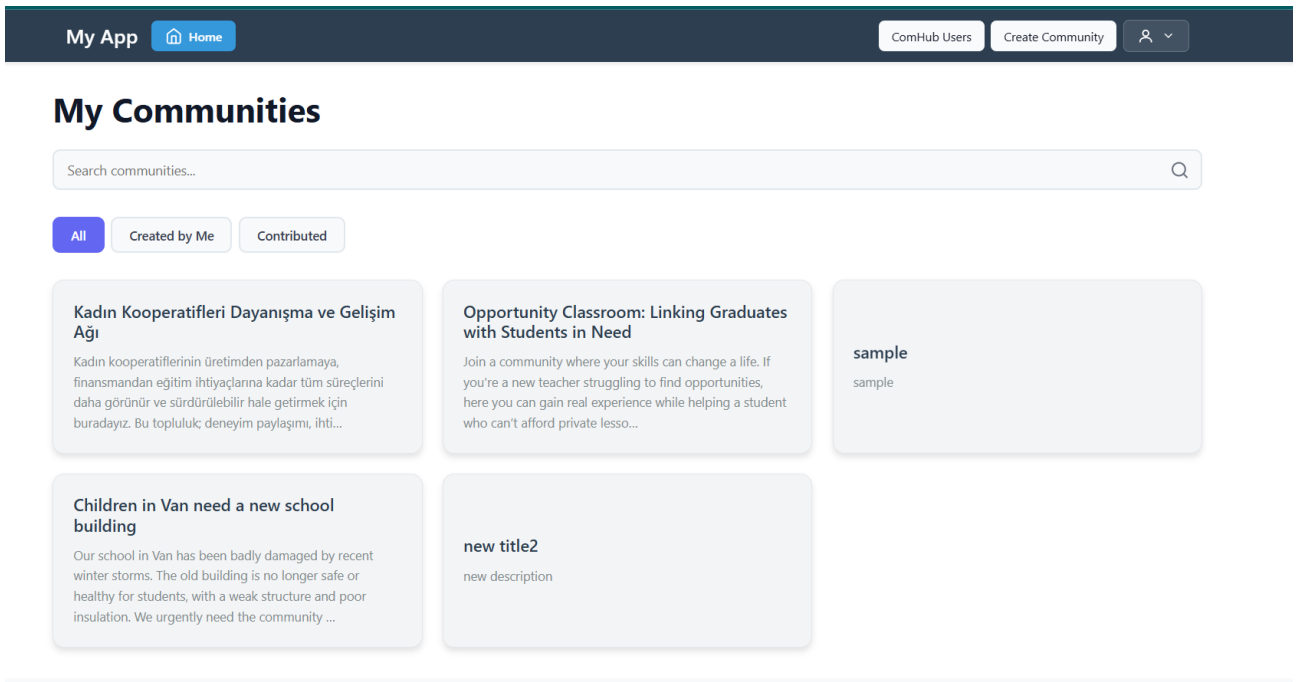
Create Community Screen 2



Community Users Screen



Other User Profile Screen



My Communities Screen

My App

Admin Dashboard

Community Hub

Dashboard

All communities

Community Users

Community Tabs

Logout

Community List

Title	Community owner	Created At	Number of tabs	Actions
Kadın Kooperatifleri Dayanışma ve Gelişim Ağı	esranrzm	13.12.2025	5	
Opportunity Classroom: Linking Graduates with Students ...	esranrzm	11.12.2025	7	
sample	emreozum	07.12.2025	2	
Children in Van need a new school building	emreozum	07.12.2025	2	
new title2	esranrzm	06.12.2025	1	

Admin Dashboard – All Communities

My App

Admin Dashboard

Community Hub

Dashboard

All communities

Community Users

Community Tabs

Logout

User List

Name	Surname	Username	Date of Birth	Occupation	Actions
admin	admin	admin	1999-12-26	admin	
digdem	cokyasar	digdemCokyasar	2011-11-11	Student	
emre	ozum	emreozum	1993-10-07	Banker	
esra	ozum	esranrzm	1999-12-26	Student	

Admin Dashboard – All Community Users

My App

Admin Dashboard

Community Hub

Dashboard

All communities

Community Users

Community Tabs

Logout

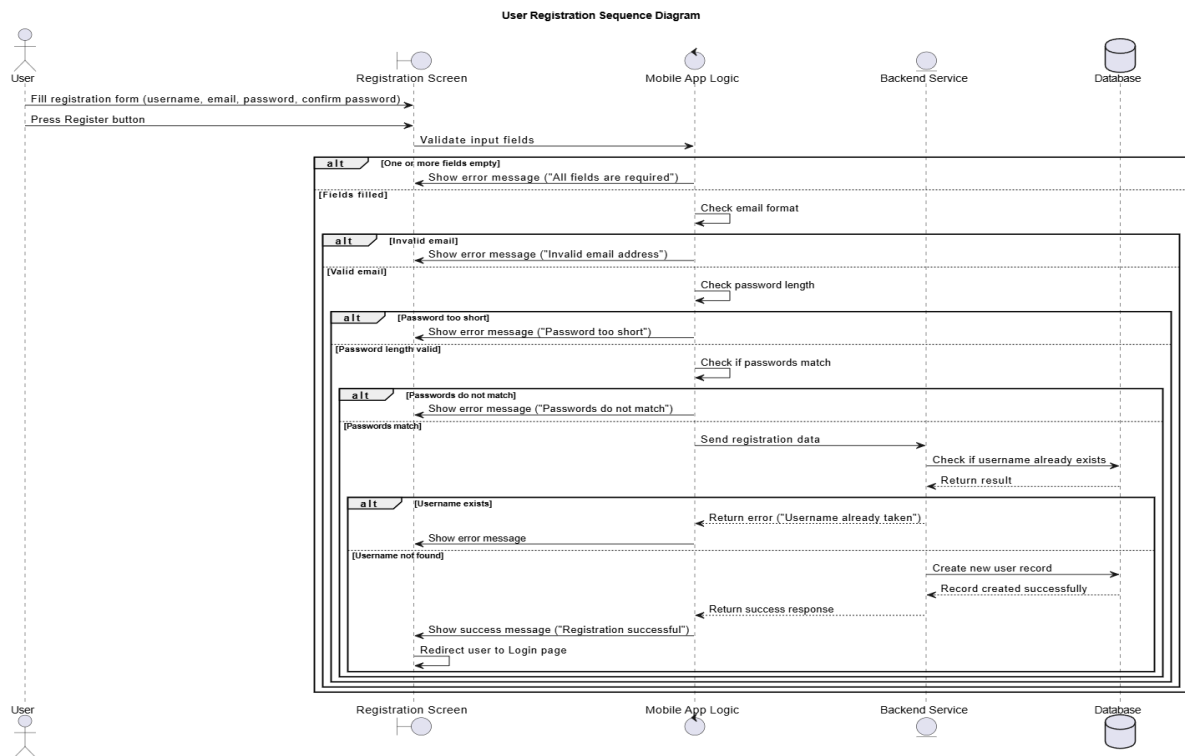
Tab List

Search tabs...

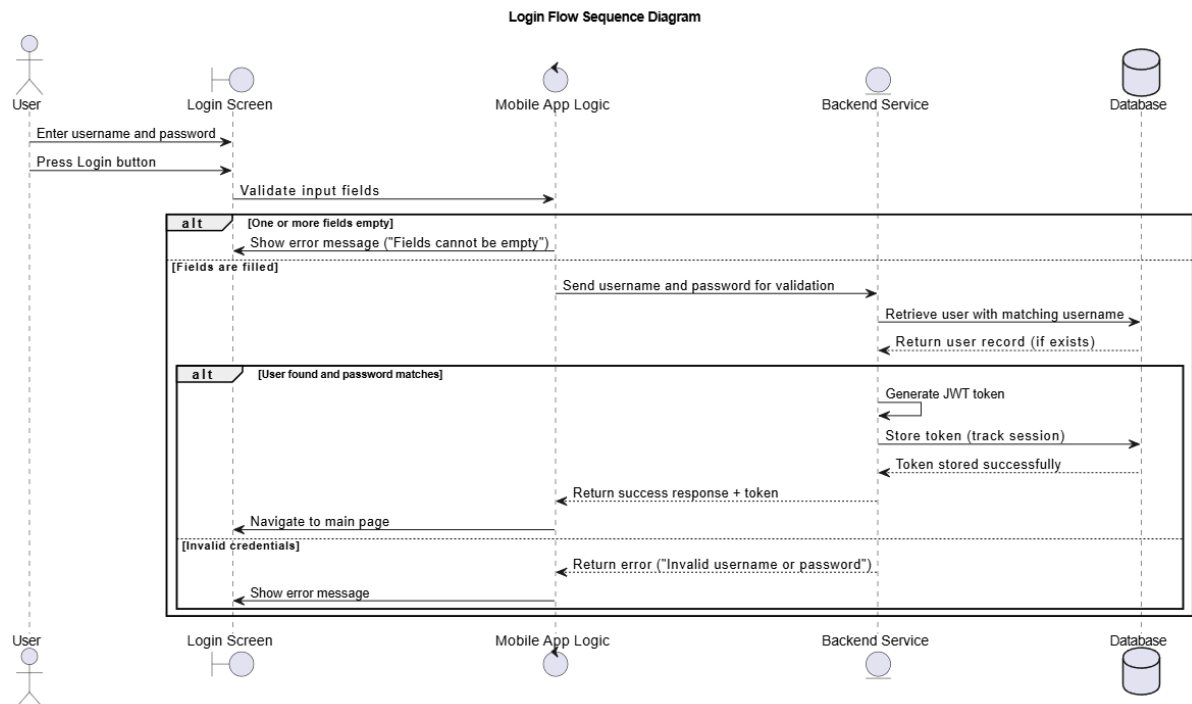
Tab Name	Community ID	Number of Inputs	Actions
sample	13	1	
Pazardaki ürünler	14	0	
Kadın kooperatiflerinde ihtiyaçlar	14	1	
Öneriler ve yorumlar	14	0	
Fuar/etkinlik katılım çağrıları	14	1	
Destek vermeye açık kooperatifler	14	1	

Admin Dashboard – All Community Hubs

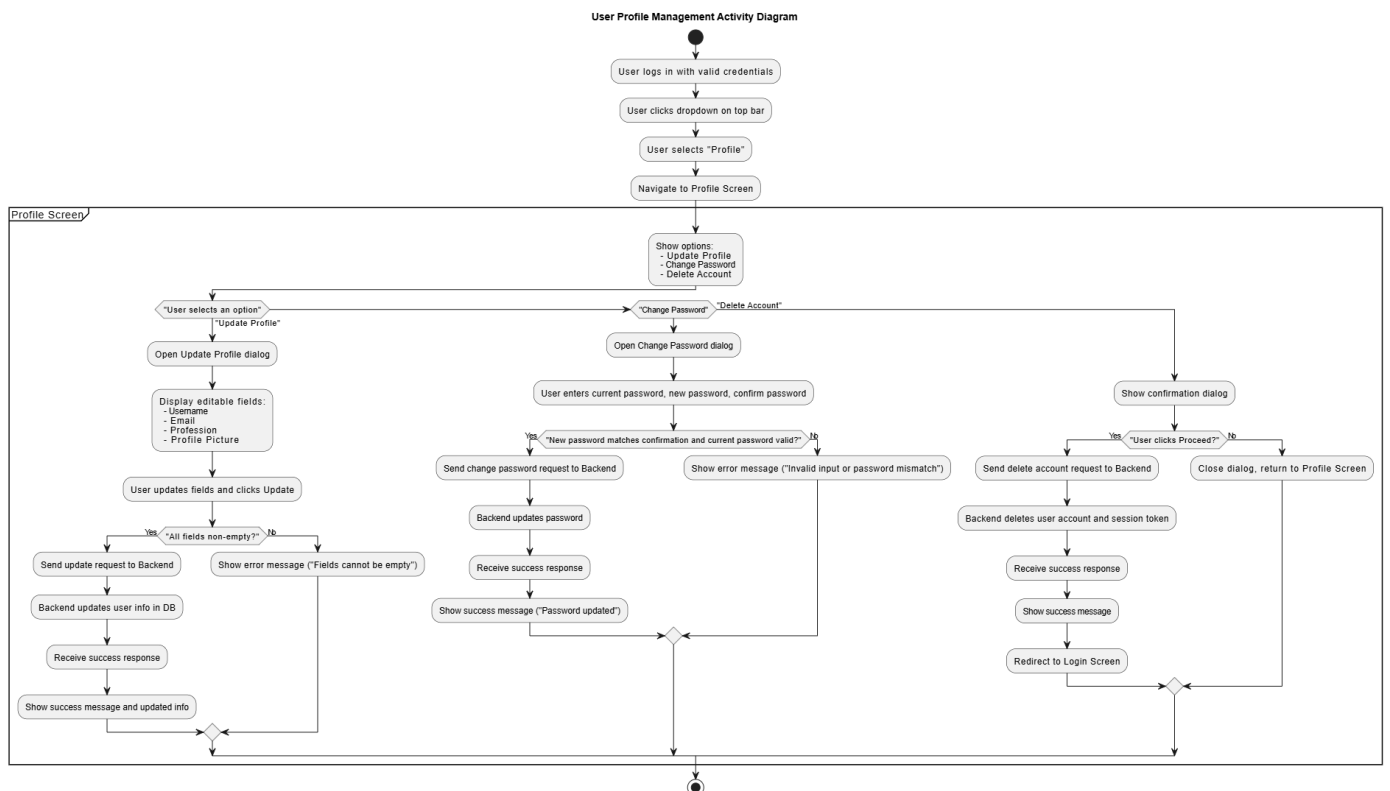
6.2 UML Diagrams



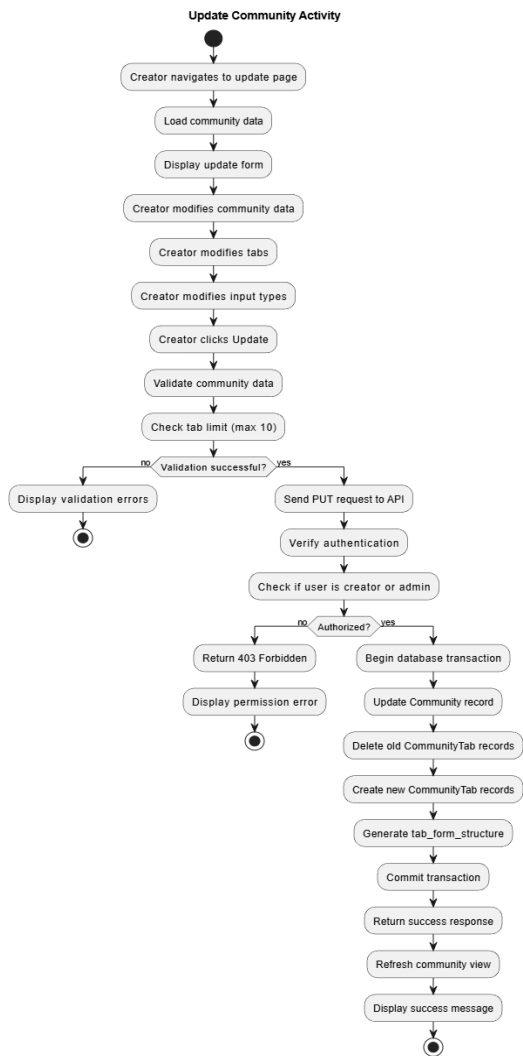
Registration Sequence Diagram



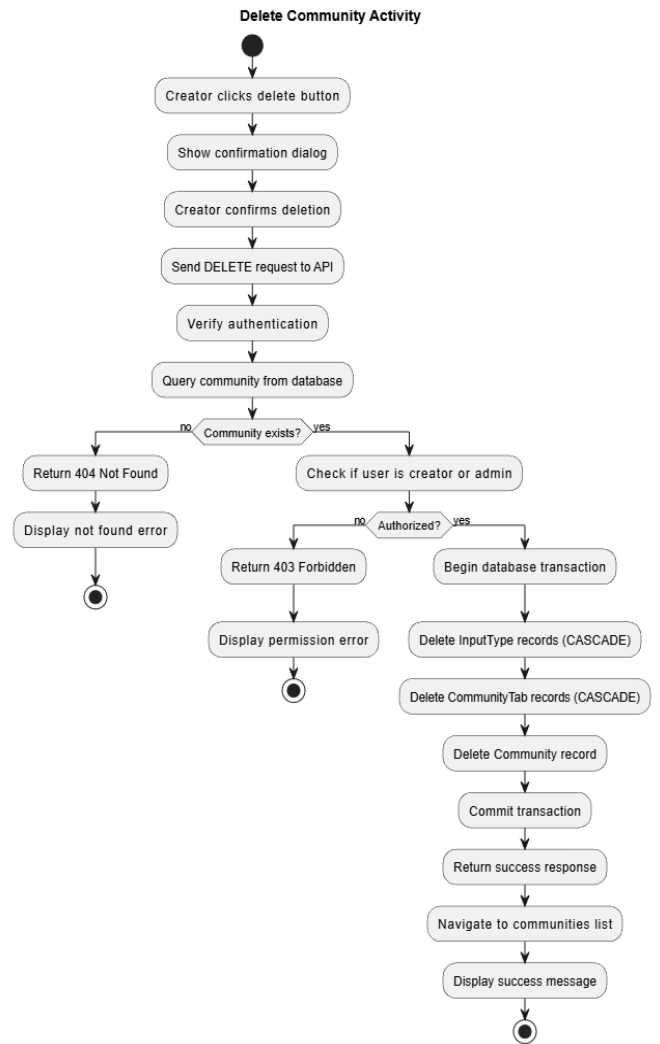
Login Sequence Diagram



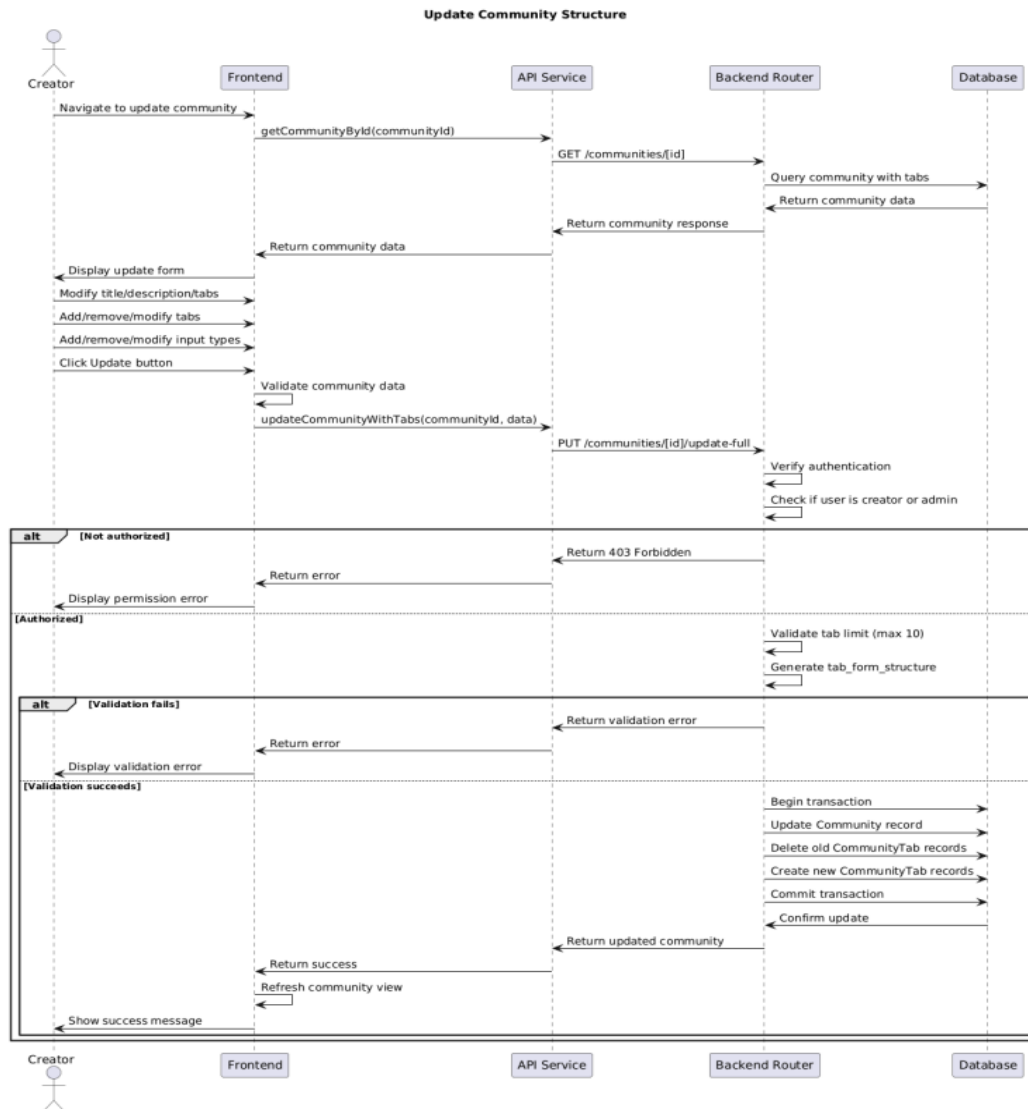
Profile Management Activity Diagram



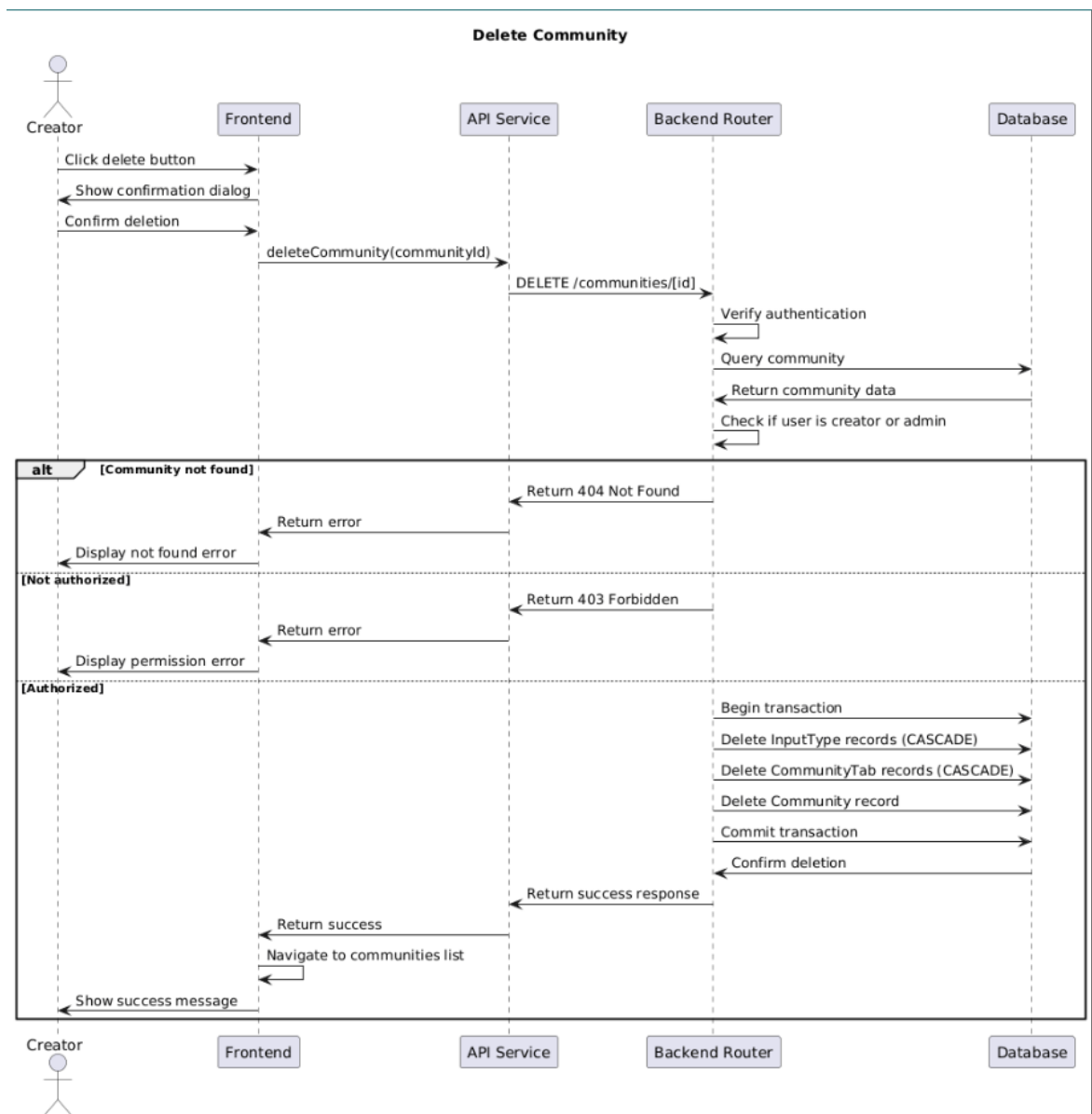
Update Community Activity Diagram



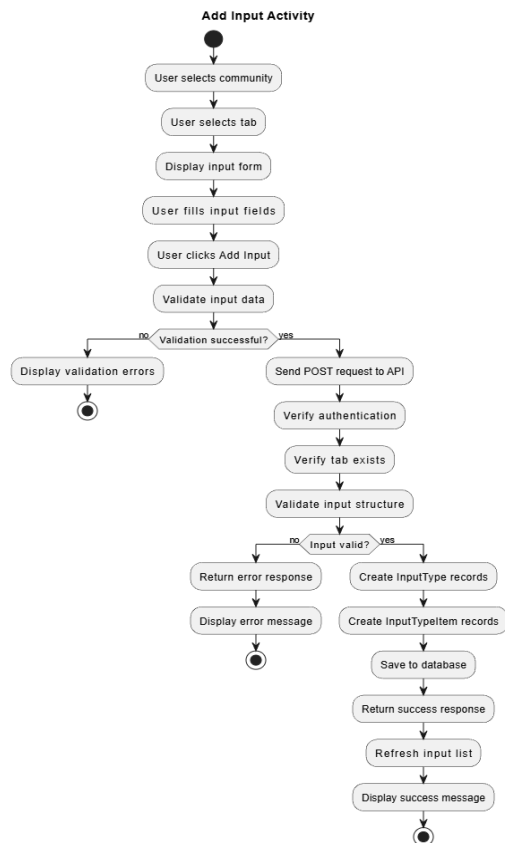
Delete Community Activity Diagram



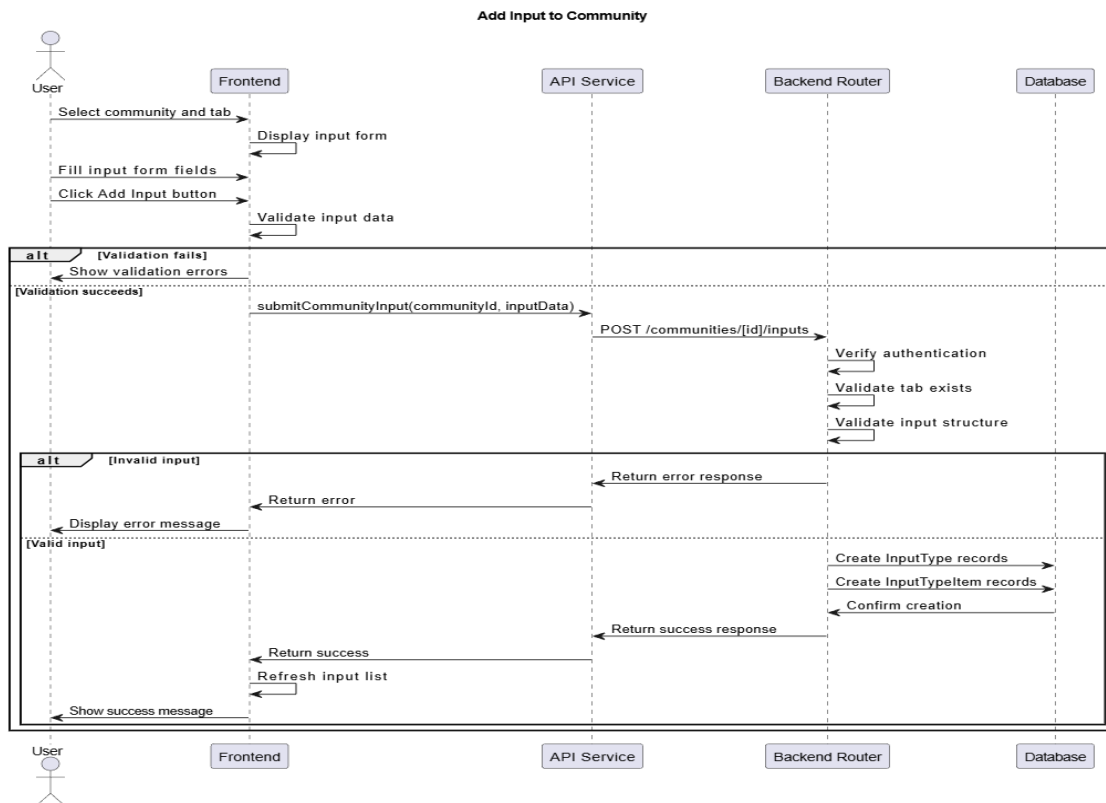
Update Community Sequence Diagram



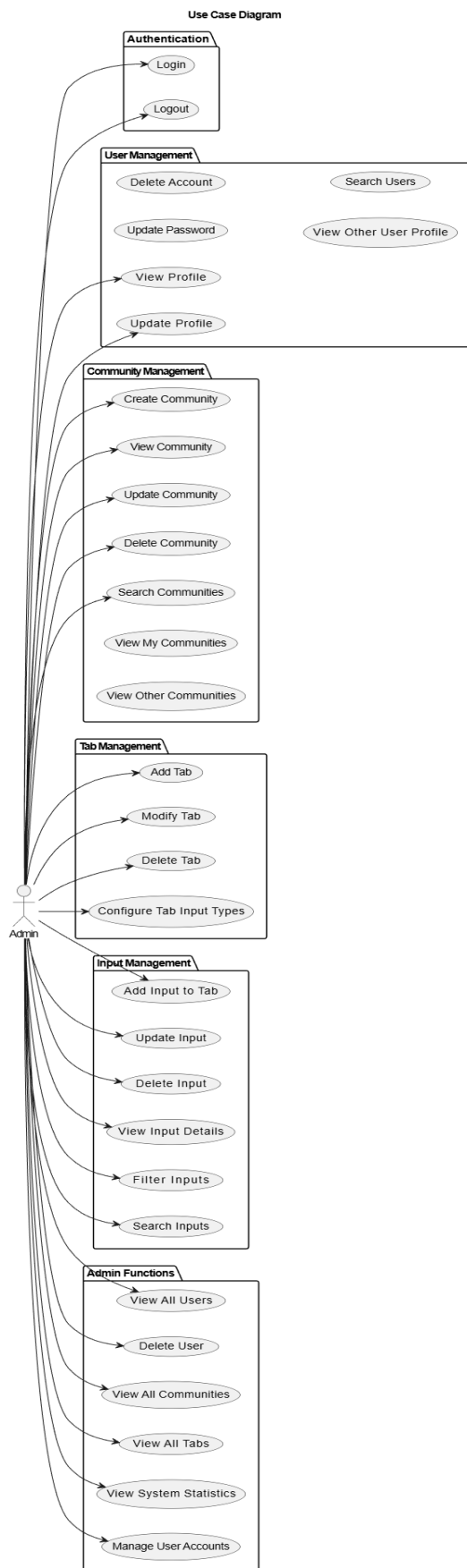
Delete Community Sequence Diagram



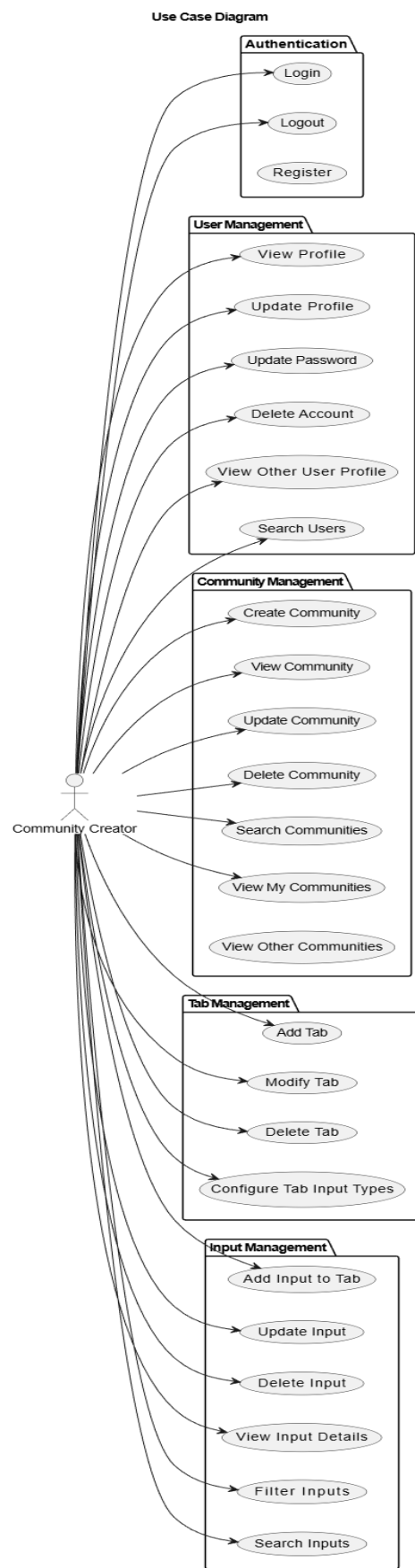
Add Input to Community Activity Diagram



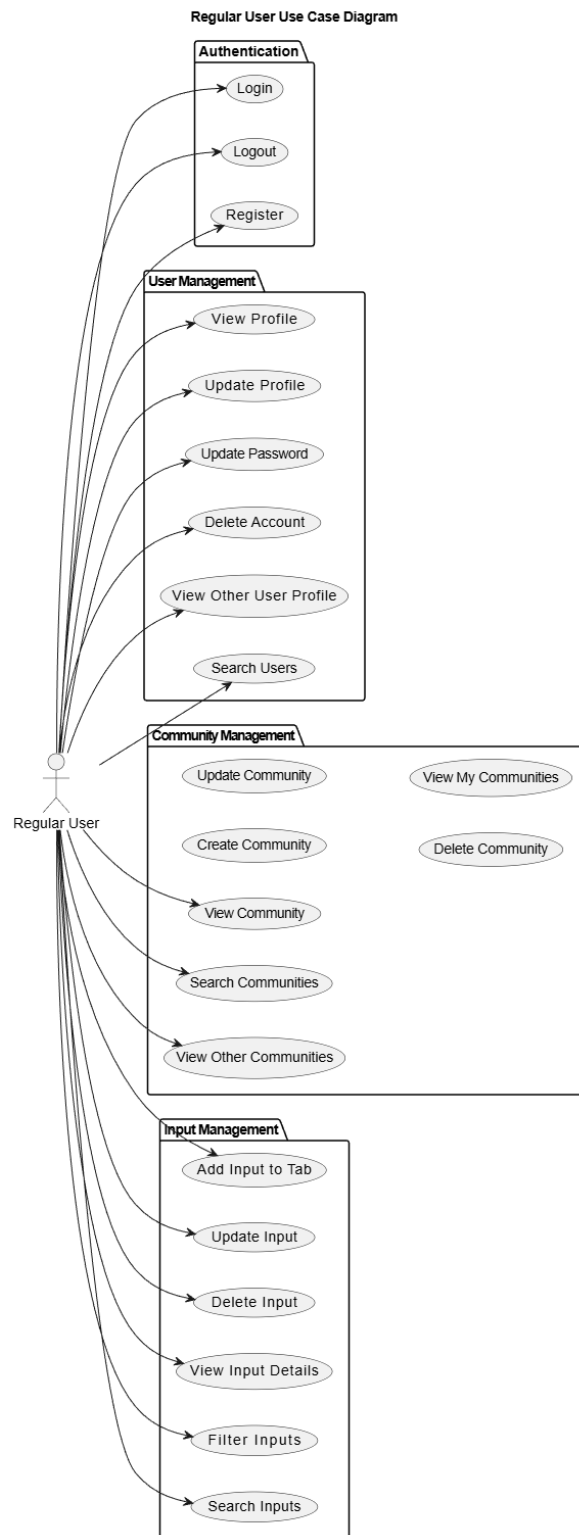
Add Input to Community Sequence Diagram



Admin User Use Case Diagram



Creator User Use Case Diagram



Regular User Use Case Diagram

7. Installation Manual

- Open your terminal or command prompt
- Navigate to the directory where you want to clone the project
- Clone the repository:
 - `Git clone ${Repository_url}`
 - `cd SWE599-Project` (if you are not already there)
- Backend Setup
 - `cd backend`
 - `python -m venv venv`
 - `venv\Scripts\Activate` (to activate the virtual environment)
 - `pip install – requirements.txt`
- Create an `.env` file under the backend directory
- Open `.env` file and fill the following information
 - `DATABASE_URL=${postgresql://username:password@localhost:5432/dbname}`
 - `SECRET_KEY=${you can generate a random secret key and put it here}`
 - `ALGORITHM=HS256`
 - `ACCESS_TOKEN_EXPIRE_HOURS=1` (you can arrange it for yourself)
- Frontend Setup
 - `cd frontend`
 - `npm install`
 - create `.env` file
 - open `.env` file and put `API_BASE_URL=${your backend url}`
 - for example, `“API_BASE_URL=http://localhost:8089/api”`
 - `npm run dev`
- Database Setup
 - Make sure PostgreSQL is running on your system
 - Connect to PostgreSQL and create a database, the database name should match with the db name that we wrote in `.env` file in backend
 - The backend will automatically create the necessary tables when you start it
- Running the Application
 - `cd backend`
 - create venv
 - activate venv
 - `uvicorn main:app --reload --port 8089`
 - open new terminal
 - `cd frontend`
 - `npm run dev`
 - go to provided url when you run `npm run dev`

8. User Manual

Project Deployed URL: <https://swe599-project-frontend.onrender.com/>

Project Users:

1. Username: emreozum, password: emre123
2. Username: esranrzm, password: esra123
3. Username: ipeq, password: ipek123

4. Username: burcuuu, password: burcu123
5. Admin username: admin, admin password: admin123 => for admin dashboard

Registration

1. Go to the url to reach the website
2. Click the "Sign Up Here!" button below the login dialog
3. You will be directed to the registration page
4. Enter the following fields:
 - a. Email Address
 - b. Name
 - c. Surname
 - d. Username
 - e. Password
 - f. Confirm Password
 - g. Profession
 - h. Date Of Birth
 - i. Consent Checkbox
5. Click the "Create Account" button below the page
6. If you entered all fields correctly your account will be created
7. You will be directed to the home page

Login

1. Go to the URL to reach the website
2. Enter your username and password
3. If the entered values are correct you will be directed to the home page
4. Your session will be started

Displaying the Profile Details

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click the Profile button on the right top corner of the page one the header
4. Dropdown will be opened
5. Click on the "Profile" button
6. You will be directed to the profile page

Displaying Other User Profile Details

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click the "ComHub Users" button on the right top corner of the page one the header
4. You will be directed to the users page
5. You can type something to search among the users

6. Or you can directly select one profile card from the list
7. Once you click on the card, you will be directed to the other user profile details page

Updating Profile Details

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click the Profile button on the right top corner of the page one the header
4. Dropdown will be opened
5. Click on the "Profile" button
6. You will be directed to the profile page
7. Click the "Edit Profile" button
8. A pop-up dialog will appear on the screen with the editable areas
9. Change the fields that you want to change
10. Click on the "Save" button

Updating Password

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click the Profile button on the right top corner of the page one the header
4. Dropdown will be opened
5. Click on the "Profile" button
6. You will be directed to the profile page
7. Click the "Update Password" button
8. A pop-up dialog will appear on the screen with three input areas
9. Enter the current password
10. Enter the new password
11. Enter the new password again for confirmation
12. Click on the "Save" button
13. Password will be updated, and you will be directed to the login page to re-login with the updated password

Deleting Profile

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click the Profile button on the right top corner of the page one the header
4. Dropdown will be opened
5. Click on the "Profile" button
6. You will be directed to the profile page
7. Click the "Delete Profile" button
8. A pop-up dialog will appear on the screen for confirmation
9. Click the "Yes" button
10. Profile will be deleted, and you will be directed to the login page

Search Communities

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. If you do not see the home page click on the “Home” button on the header at the left top corner of the page
4. In the home page click on the search box at the top of the page
5. Type whatever you want to search for
6. The matched results will be seen automatically once you write something
7. Select one of the results to see the details of it

Displaying Community Details

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. If you do not see the home page click on the “Home” button on the header at the left top corner of the page
4. In the home page click one of the community cards that you want to explore more
5. You will be directed to the community details page
6. At the beginning of the page, you will see the title and the description. Right next to it you will see the community creator details. Below the description you will see the tabs that the community has. Under each tab you can see the added inputs as an input. You can search among them to find what you need.

Creating a new Community

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click on the “Create Community” button on the header
4. You will be directed to the community creation page
5. You need to enter title and description of the community you wanted to create
6. After entering the values, you will see tab creation area under the description box
7. If you want to add some tabs and construct forms inside of it you need to click on the “Create New Tab” button
8. Tab details section will be opened once the button is clicked
9. Fill all the details about the tab you want to create
 - a. Choose a color from color palette
 - b. Enter the name of the tab
 - c. Enter the description of the tab
10. Below the tab description section, you will see form area
11. You can add as much input as you want
 - a. Write the name/title of the input
 - b. Select the type of the input (free-text, dropdown, multiselect, url, date, location)
 - c. If your selection is multiselect or dropdown, you will see a new area for adding input fields. These section you will be able to add dropdown/multiple selection items
 - d. Write the option you want to add and click the “Add Item” button
 - e. Once you add the item a list will appear next to the “Add Item” button. You can see the added items in that list

- f. Once you are done with the input field adding, click the “Save Field” button below the input field adding section
12. Once you are done with the input adding, you can add another tab with the same “Create New Tab” button
13. Once you are done with everything, scroll down and press on the “Create” button.
14. The community will be created
15. You will be directed to the community details page

Updating the owned community

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click the Profile button on the right top corner of the page one the header
4. Dropdown will be opened
5. Click on the “My Communities” button
6. You will be directed to the page where you can see the communities that you created, you added input, or all of them
7. Click on the “Created by Me” button
8. Click one of the community cards on the screen under the category “Created by Me”
9. You will be directed to the community details page
10. Click on the “Update Community Details” button on the screen
11. You will be directed to a new page where you can edit everything on the screen
12. Once you are done with your updates, click on the “Update Community” button
13. You will see a success message and you will be directed to the community details page

Deleting the owned community

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. Click the Profile button on the right top corner of the page one the header
4. Dropdown will be opened
5. Click on the “My Communities” button
6. You will be directed to the page where you can see the communities that you created, you added input, or all of them
7. Click on the “Created by Me” button
8. Click one of the community cards on the screen under the category “Created by Me”
9. You will be directed to the community details page
10. Click on the “Delete Community” button on the screen
11. A confirmation pop-up dialog will be appeared on the screen to ask for the last confirmation
12. Click the “Yes” button
13. The community and inputs added to those communities will be deleted
14. You will be directed to the My Communities page. (Or the page where you came from)

Adding New Input to the Community

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. If you do not see the home page click on the “Home” button on the header at the left top corner of the page
4. In the home page click one of the community cards that you want to explore more
5. You will be directed to the community details page
6. Scroll down the page and click on the “Add Input to tab \${tab_name}” button for the selected tab
7. A pop-up dialog will appear on the screen that contains the form that you can fill to create a new input
8. Fill all mandatory fields
9. Once you are done with them, click the “Add” button to save the input
10. Your input will be added, and you will see the updated Community Inputs with your input

Updating the Input that you previously added to the Community

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. If you do not see the home page click on the “Home” button on the header at the left top corner of the page
4. In the home page click one of the community cards that you want to explore more, and you have added an input before
5. You will be directed to the community details page
6. Scroll down the page and find the input that you want to update
7. Click on the edit icon for the input you want to update
8. A pop-up dialog will appear on the screen that contains the form that you filled before
9. Update your answers
10. Click on the “Update” button on the form
11. Your answers will be updated

Deleting the Input that you previously added to the Community

1. Login to the system or register if you do not have any account following the mentioned steps
2. You will be navigated to the home page
3. If you do not see the home page click on the “Home” button on the header at the left top corner of the page
4. In the home page click one of the community cards that you want to explore more, and you have added an input before
5. You will be directed to the community details page
6. Scroll down the page and find the input that you want to update
7. Click on the delete icon for the input you want to delete
8. A pop-up dialog will appear on the screen for confirmation
9. Click on the “Delete” button
10. Your input will be deleted

Accessing the Admin Dashboard

1. Open web application
2. Enter admin credentials in login screen
3. You will be directed to the admin dashboard
4. You can select all communities tab to see the list of communities that is actively exist in the system
5. You can select all users to see all the user accounts
6. You can select all tabs to see the created tabs
7. To see the details of each record you can click the eye icon at the end of the table row
8. Once you click the eye icon button, you will be able to see the detail of selected object
 - a. If it is a user, you will be directed to the other user profile page
 - b. If it is a community, you will be directed to the community details page
 - c. If it is a tab, you will be directed to the community details and the tab will be selected

Deleting community/tab/user in the admin dashboard

1. Open web application
2. Enter admin credentials in login screen
3. You will be directed to the admin dashboard
4. Select community/tab/user option from the side bar
5. You will see delete option for the community and the user records
6. You can directly delete it in admin dashboard
7. For deleting the tab, you need to go to the community details and click the button "Update Community" button
 - a. You will be directed to the community update page
 - b. You can scroll down to find the tab you want to delete and click "x" button to delete it from the community

9. Test Results

1. Tests for authentication and user management endpoints

TEST NAME	TEST DESCRIPTION	TEST RESULT
Successful Registration	Test successful user registration	Passed
Unsuccessful Registration	Test registration with duplicate username	Passed
Unsuccessful Registration	Test registration with duplicate email	Passed
Unsuccessful Registration	Test registration with invalid email format	Passed
Unsuccessful Registration	Test registration with password shorter than 6 characters	Passed
Successful Login	Test successful login	Passed
Unsuccessful Login	Test login with non-existent username	Passed

Unsuccessful Login	Test login with incorrect password	Passed
Reading Profile Information	Test getting current user information	Passed
Updating Profile Information	Test updating user profile	Passed
Unsuccessful Updating Profile Information	Test updating profile with duplicate username	Passed
Updating Password	Test updating user password	Passed
Unsuccessful Updating Password	Test updating password with wrong current password	Passed
Deleting Profile	Test deleting user account	Passed
Getting all users	Test getting all users	Passed
Get user by name	Test getting users by name	Passed
Get user by username	Test getting user by username	Passed
User not found	Test getting non-existent user by username	Passed
logout	Test successful logout	Passed
Failed logout	Test logging out twice with same token	Passed

2. Tests for community related endpoints

TEST NAME	TEST DESCRIPTION	TEST RESULT
Successful community creation	Test successful community creation	Passed
Successful community creation	Test creating community without tabs	Passed
Unsuccessful community creation	Test creating community without authentication	Passed
Getting all communities	Test getting all communities	Passed
Getting all communities by id	Test getting community by ID	Passed
Unsuccessful getting communities	Test getting non-existent community	Passed
Successful getting created communities	Test getting current user's created communities	Passed
Successful getting other created communities	Test getting communities created by others	Passed
Successful getting communities by id	Test getting communities by user ID	Passed
Successful updating communities	Test updating community	Passed
Successful updating communities with tabs	Test full community update with tabs	Passed
Successful deleting communities	Test deleting community	Passed
Unsuccessful updating communities	Test deleting community as non-creator	Passed
Successful adding community input	Test submitting input to a community	Passed

Successful getting communities input count	Test getting count of community inputs	Passed
Successful getting community input	Test getting community inputs	Passed
Successful updating community inputs	Test updating a community input	Passed
Successful deleting community input	Test deleting a community input	Passed
Successful getting cities	Test getting cities for a country	Passed

3. Tests for admin related endpoints

TEST NAME	TEST DESCRIPTION	TEST RESULT
Successful getting admin statistics	Test getting admin statistics	Passed
Successful getting all users	Test getting all users as admin	Passed
Unsuccessful getting all users	Test getting all users as non-admin	Passed
Successful deleting a user	Test deleting a user as admin	Passed
Unsuccessful deleting admin user	Test admin trying to delete their own account	Passed
Unsuccessful deleting a user	Test deleting user as non-admin	Passed
Successful getting all communities	Test getting all communities as admin	Passed
Unsuccessful getting all users	Test getting all communities as non-admin	Passed
Successful getting all tabs	Test getting all tabs as admin	Passed
Unsuccessful getting all communities	Test getting all tabs as non-admin	Passed

10. Future Work

The improvements and additional features that can be done in later stages can be summarized as follows. At the beginning of the project, while designing it, I planned to add AI tools to the project as a kind of “summarizer,” since these tools are now being used in many areas. However, later on, because there were other features that needed to be prioritized and this one was not among the most important, I decided not to include it in the final demo. The purpose of this feature was to show the user, in the form of a summary, what was happening in the community, what kinds of answers people were giving, which tab was being used the most, and what the overall structure looked like at the moment the summary was created. I first saw a similar feature in the comments section of the Trendyol application and thought it could also fit this project. But for the reasons mentioned above, I classified it as “nice to have” and planned to add it in later stages.

Another future work could be improving the structure I currently use in the storage phase of the input-gathering mechanism. While designing the project and creating the designs, I decided to store the data using a JSON structure, and so far this has worked successfully. During the feedback meeting, my instructor suggested another structure — JSON-LD, which can be seen as a more advanced version of JSON. Using JSON-LD in later stages could be a good option to make the storage

part and information-management processes easier. When I researched it in more detail, I found the following information about JSON-LD.

In the current structure of the project, the JSON format makes it possible to store data in a structured way. It is especially useful when communicating with the API, since it is used as the request and response format and makes it easier to send and receive data in an organized form. In the project, I made extensive use of JSON while building the user inputs, the forms under the tabs, and storing them. JSON-LD, also known as JSON Linked Data, is a more customized and advanced version of JSON. It is used by advanced search engines like Google and Bing, and thanks to its structure, it allows machines and algorithms to understand the meaning carried by the data. It has two special vocabularies called “@context” and “@type.” In this sense, “context” tells the machine which vocabulary to use, while “type” provides information about what the data is. Since it follows W3C standards, this structure improves the areas where JSON remains limited and helps us have a more structured, meaning-carrying data format. This feature will become especially useful when the AI summarization tool is added, because by using JSON-LD, which contains meaningful data, it will be possible to get more high-quality and informative answers from the AI tool.

11. References

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